

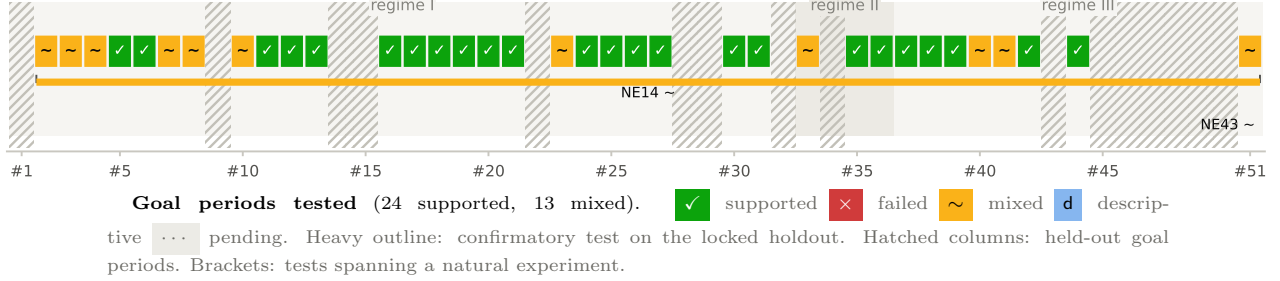
H50: Field or coupling: transfer functions and lag tell which

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2.5/5 Usefulness 2.5/5

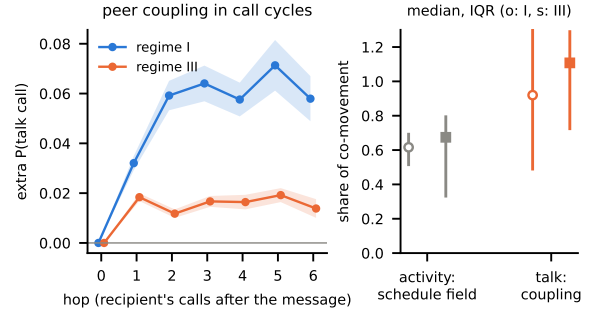
Bottom line. Activity co-movement is a field set by the scheduler; talk spreads by a coupling gated at one read-out call, and agents rarely relay it further.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The question is whether agents co-move because a common input hits everyone (a field) or because they read each other (a coupling). In call cycles, a field moves every agent at its next call. A coupling acts only at the recipient's first call that can read the message. The DQ1 context ledger gives a placebo-corrected discontinuity, hop by hop: calls that start just after a message against calls just before it. The analysis also fits input impulse responses and decomposes co-movement on 71 non-holdout units (35 periods). **Activity co-movement is the schedule:** day start and stop explain about 0.6 of it beyond a null. **Talk co-movement is a gated coupling.** A peer message raises the chance that the next call is a talk call (45/71 units, none negative), always at hop 1. It accounts for roughly all talk co-movement. HH167's regime split fails. The holdout is not run.



Observables. Left: excess talk probability of a recipient's call, by hop after a peer message (IVW pooled; 95% band). The response is zero at the in-flight call (hop 0) and steps up at the read-out call (hop 1). Regime III (13-s chained calls) drops at hop 2, then plateaus. Regime I (scheduled chat calls) keeps rising. Right: share of activity co-movement explained by the schedule edges (gray), and of talk co-movement by the gated coupling (orange; > 1 is overshoot).

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

PHYSICAL MODEL

Gated linear response (models 02, 01). Agent i makes calls c at times t_c . A message posted at t_m enters the context at the first call with $t_c > t_m$ (hop 1). The call outcome Y_c (talk, work, idle) responds to inputs u_{c-k} (bookends, human messages, nudges, platform errors). It also responds to peer messages m_{c-k} first read k hops earlier. s is the call start minus the message time. e_{it} is agent-day-demeaned minute activity or talk. S is the equal-time co-movement.

$$\text{logit } P(Y_c=1) = h_i + \sum_k g(k) u_{c-k} + \sum_k J(k) m_{c-k}$$

$$J_k = E[Y | s \rightarrow 0^+] - E[Y | s \rightarrow 0^-] - \text{placebo}, \quad \delta(h) = \sum_{k \leq h} J_k$$

$$S = \sum_t [(\sum_i e_{it})^2 - \sum_i e_{it}^2] / \sum_t \sum_i e_{it}^2$$

H50-field-vs-coupling-transfer-lag · generated 2026-10-04 11:54 by infra/summaries/build_summaries.py · card hypotheses/H50-field-vs-coupling-transfer-lag/README.md · ratings: H50 agent (suggested)

RESULTS VS PREDICTIONS

Prediction (dated)	Observed	Verdict
P1 synthetic recovery	J_1 bias -2 to -17% (III); dead time 2 $\rightarrow$ onset hop 3; nulls 3/35	passed
P2 inputs at read-out	in-room human $J_1 > 0$, hop 1; nudged agents talk at hop 1 (0.14), not later	mixed
P3 low-pass, corner 5–40 min	coherence $8-10\times$ at 15–60 min; corner ill-defined (undershoot)	mixed
P4 field: zero relative lag	onsets IQR 9–22 s (< 2 cycles); #38 agents halted at pause	supported
P5 gated coupling, hop 1	$J_1 > 0$ in 45/71, < 0 in 0; onset hop 1; no decay in regime I	mostly
P6 decomposition	activity field 0.55 (III) $\checkmark$, 0.51 (I, pred. ≤ 0.2) $\times$; talk coupling $> 57/68$	mostly
HH167 regime split	regime-I calls not message-triggered (ratio 1.03–1.07)	rejected
NE43 switch-offs	nudge path gone; coupling $\times 0.69$; edge field persists	mixed
G51 human relay	plain human messages: hop-1 field; relay underpowered	mixed
G38 pause gates	82% start before any peer talks; 94% never call after pause	supported

GOAL PERIODS IN DETAIL

G02 $\sim$ mixed
G03 $\sim$ mixed
G04 $\sim$ mixed
G05 $\checkmark$ supported
G06 $\checkmark$ supported
G07 $\sim$ mixed
G08 $\sim$ mixed
G10 $\sim$ mixed
G11 $\checkmark$ supported
G12 $\checkmark$ supported
+27 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	2
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

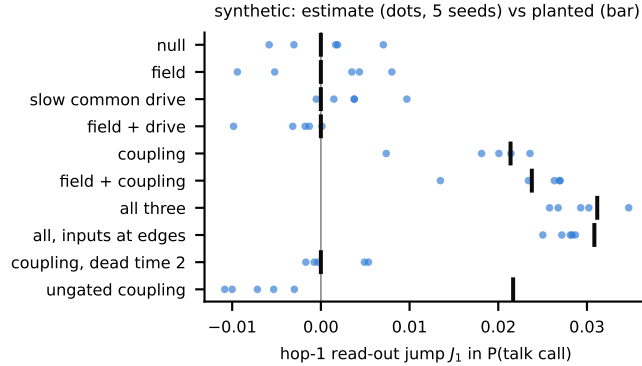
CAVEATS

Regime-I call starts are latency-placed. The jump survives on logged-start recipients (#24–#31; 0.030 vs 0.034), but early regime I is unchecked. The rising regime-I kernel beyond hop 2 has no synthetic validation. Per-unit CIs are mildly anti-conservative (synthetic false positives about 9%). Decomposition shares are model-based and do not add up, because coupling amplifies fields. $f_C > 1$ is overshoot at low talk co-movement. Room changes and roster joins confound NE43. The #51 target and relay tests have about 100 pairs. No content channel exists yet. The holdout is not run.

ROUND 2

Add a content gate with a matched-age control. Validate the regime-I kernel. Compare per-agent “impedance” with H29/H32 influence. Confirm on the holdout, including NE23 (nudger off and on).

- **Essence**. which levers move a swarm together (fields) and which only propagate through who reads whom (couplings), measured in the units the scaffold imposes (calls), so an operator knows what an input will do to collective behavior.
- **H50-R1** Content gate (C3).
- **H50-R2** Validate the regime-I dead-time kernel.
- **H50-R3** HH188 frequency-dependent T_{eff}.
- **H50-R4** HH190 agent impedance.
- **H50-R5** Human-input relay with power.
- **H50-R6** HH187 graph clause.
- **H50-R7** Confirm on the holdout (confirm.py: CP1–CP5, NE23 nudger off/on).



Second figure. Synthetic swarms at village sampling with real input schedules (5 seeds each). The read-out jump finds planted coupling with or without fields, slow common drives or edge-clustered inputs. It reads zero for fields and drives. It recovers a 2-hop dead time as a later onset. It goes negative for ungated coupling.